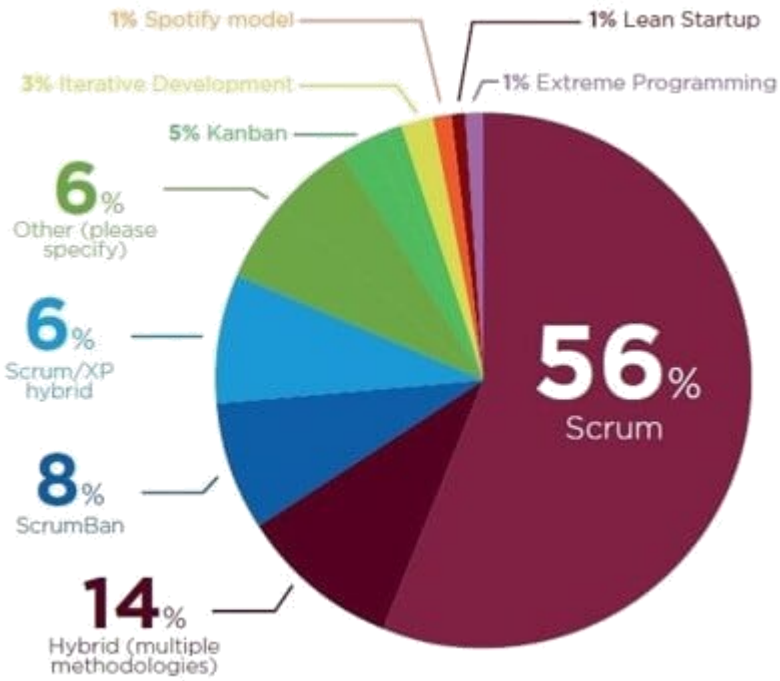
Scrum



Capstone Project 2025/26

Maura Cerioli

We're losing the relay race

"The… 'relay race' approach to product development…may conflict with the goals of maximum speed and flexibility. Instead a holistic or 'rugby' approach—where a team tries to go the distance as a unit, passing the ball back and forth—may better serve today's competitive requirements."

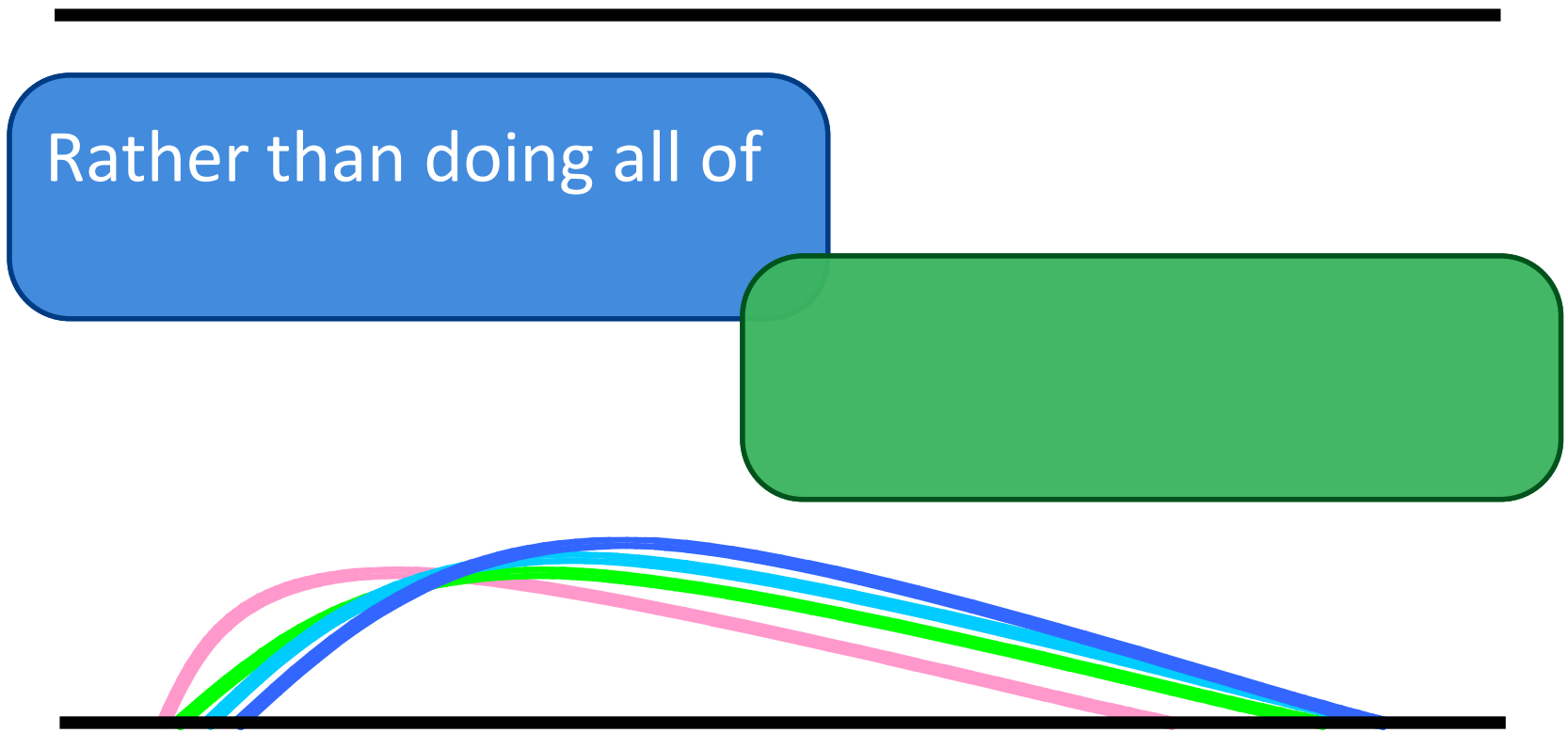
Hirotaka Takeuchi and Ikujiro Nonaka, "The New New Product Development Game", *Harvard Business Review*, January 1986.





Dibris

Sequential vs. overlapping development



Source: "The New New Product Development Game" by Takeuchi and Nonaka. *Harvard Business Review,* January 1986.

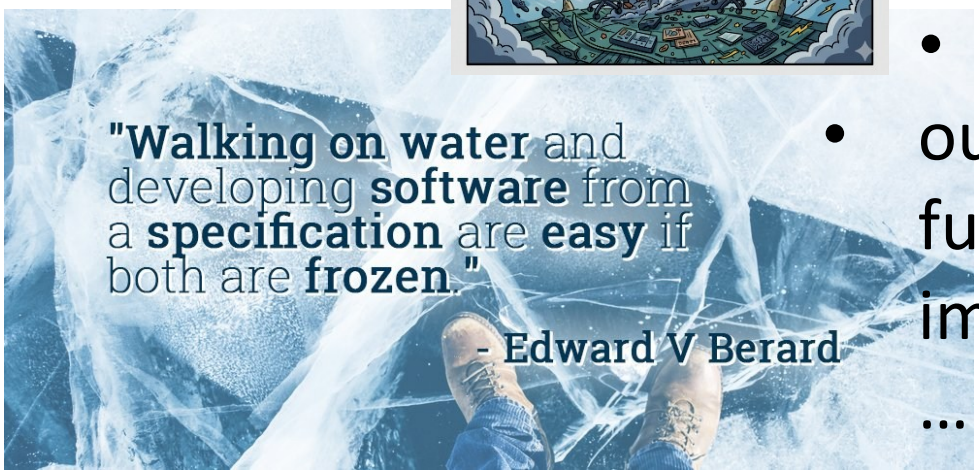


Sprints = iterations

Dichotomy: embracing changes vs. need for stable requirement to get better productivity



- input: **frozen** requirements
  - **only** the sprint drivers
  - including DoD
- time boxed
  - in time: limited by requirements' instability (2 weeks – 1 month)
  - in effort: constant to get a better rhythm
  - risk containement
- output: fully tested correct implementation … or failure

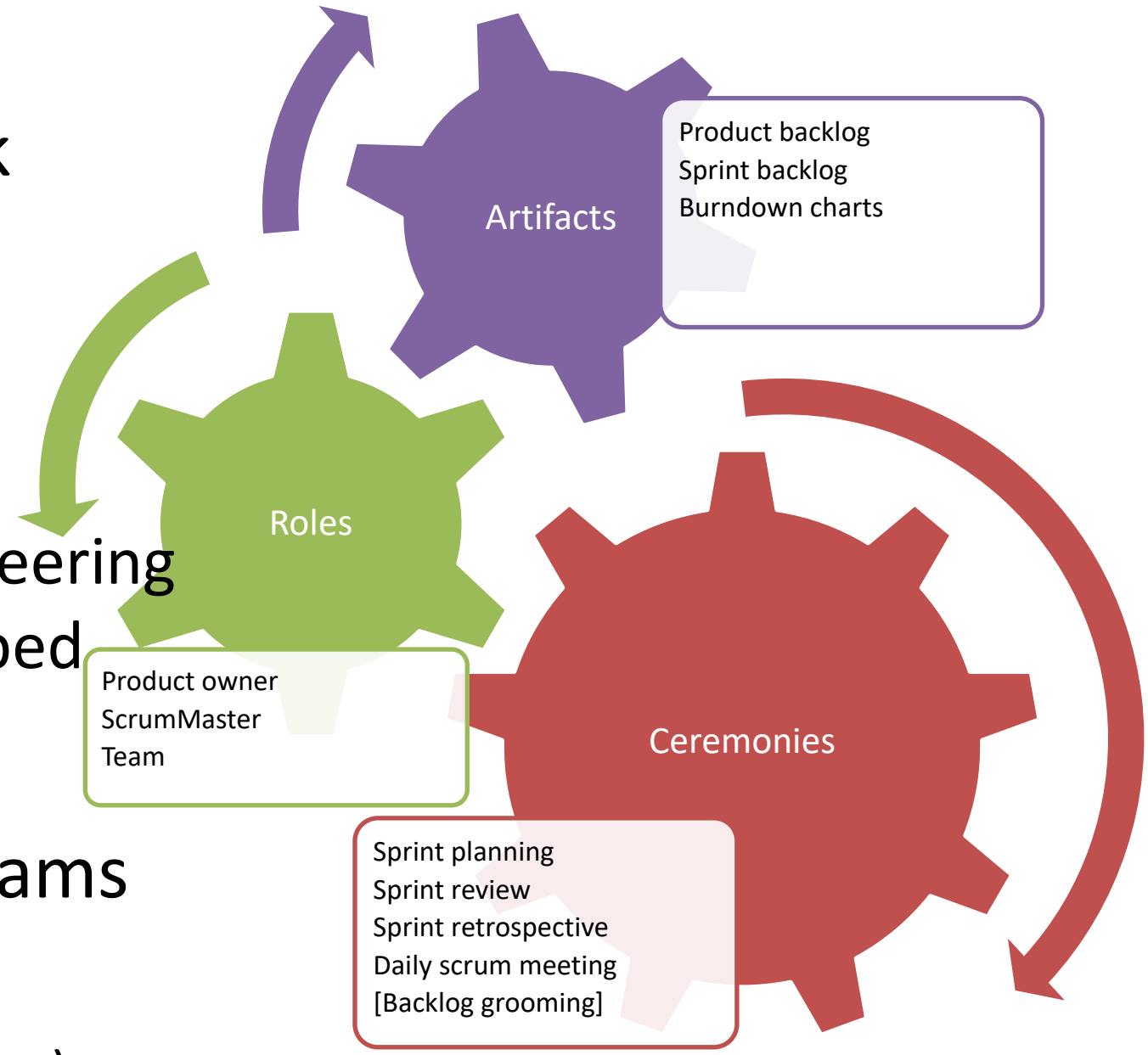




Dibris

Other Main Characteristics

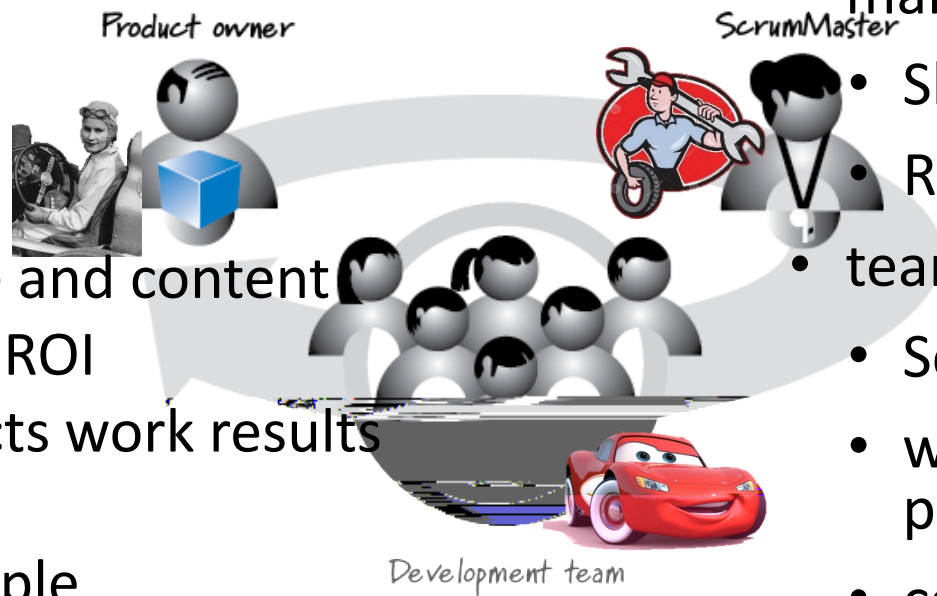
- Scrum framework
  - 3 Roles
  - 4/5 Ceremonies
  - 3 Artifacts
  - No specific engineering practices prescribed
- Self-organizing teams
  - Generative rules
    - (**not** inclusive rules)



Scrum **Key** Roles

Proxy for business stakeholders

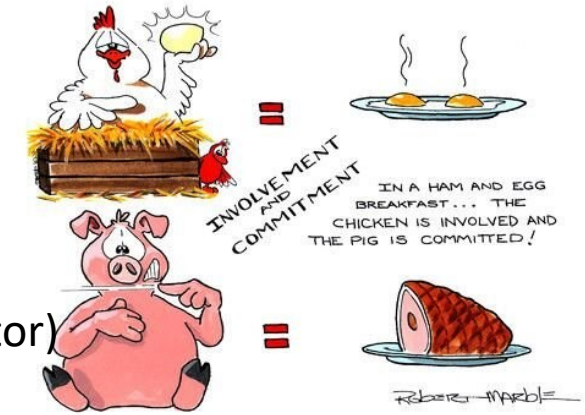
- Decides
  - features
  - priorities
  - release date and content
- Responsible for ROI
- Accepts or rejects work results



Interface team ⇔ management

- management side
  - Shield from interferences
  - Removes impediments
- team side
  - Scrum coach
  - watches for/quenches problems
  - cooperation facilitator

- 5-9 people
- Cross-functional:
  - Programmers, testers, user experience designers, etc.
- Self-organizing
- Membership
  - changes **only** between sprints
  - full-time (but for experts, e.g., database administrator)

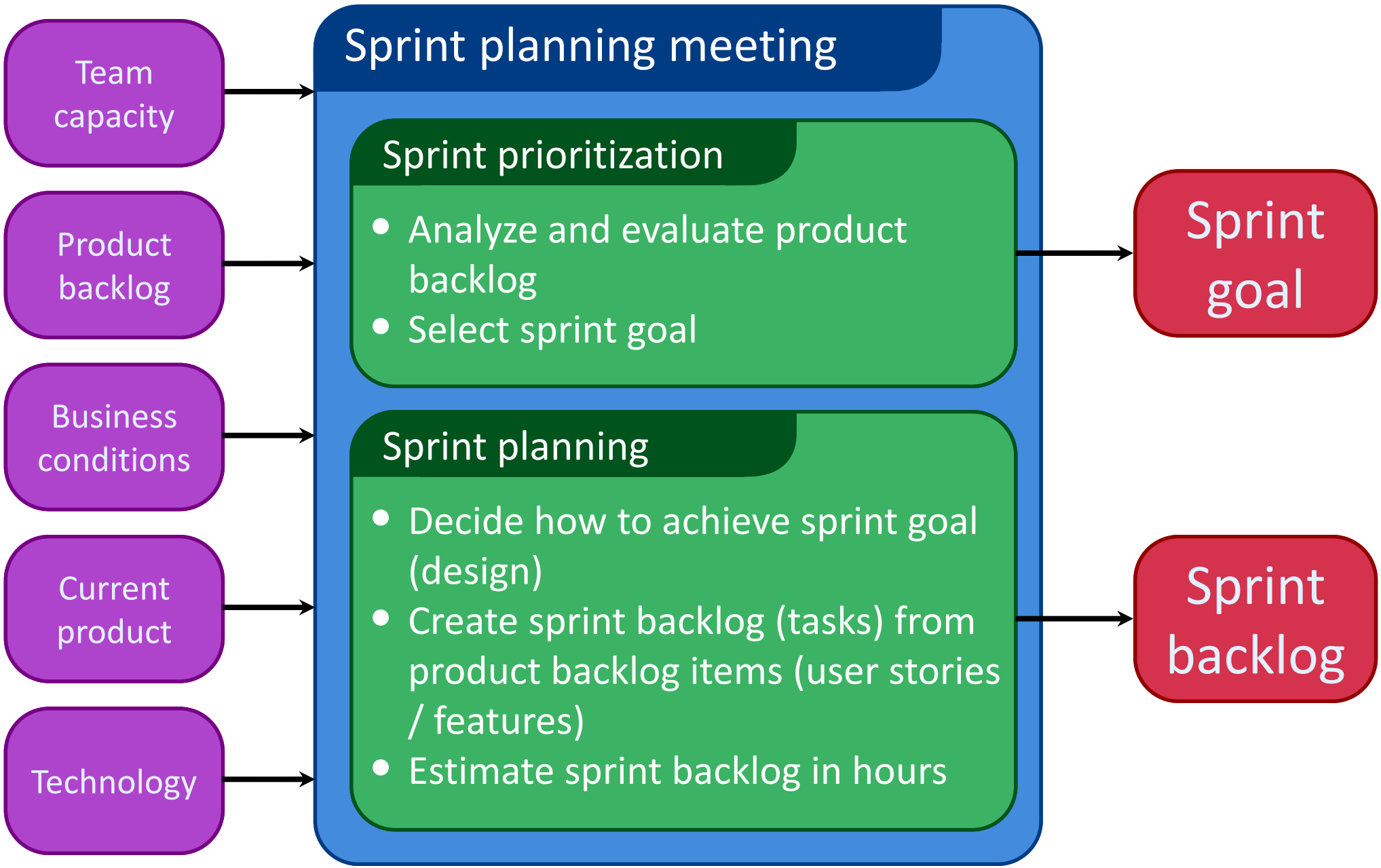


Overall Scrum Process

Artifact 1: Product backlog
A **totally ordered** list of
- user needs
- technical issues
- technical debts

Artifact 2: Sprint backlog
A **collection** of tasks

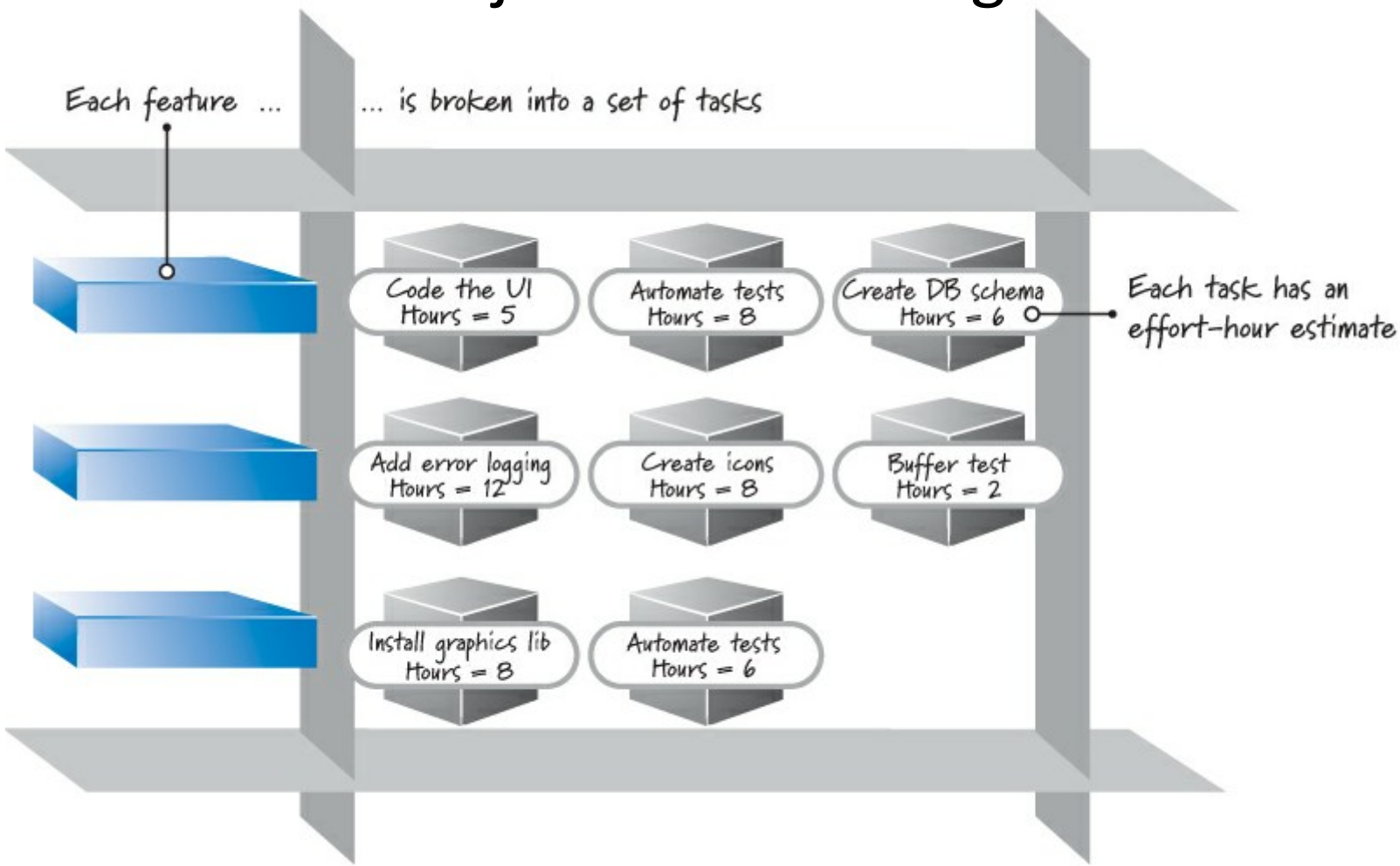



Team capacity
Product backlog
Business conditions
Current product
Technology
Sprint planning meeting
Sprint prioritization
Analyze and evaluate product backlog
Select sprint goal
Sprint planning
Decide how to achieve sprint goal (design)
Create sprint backlog (tasks) from product backlog items (user stories / features)
Estimate sprint backlog in hours
Sprint goal
Sprint backlog

Dibris

Sprint Backlog

just in time design



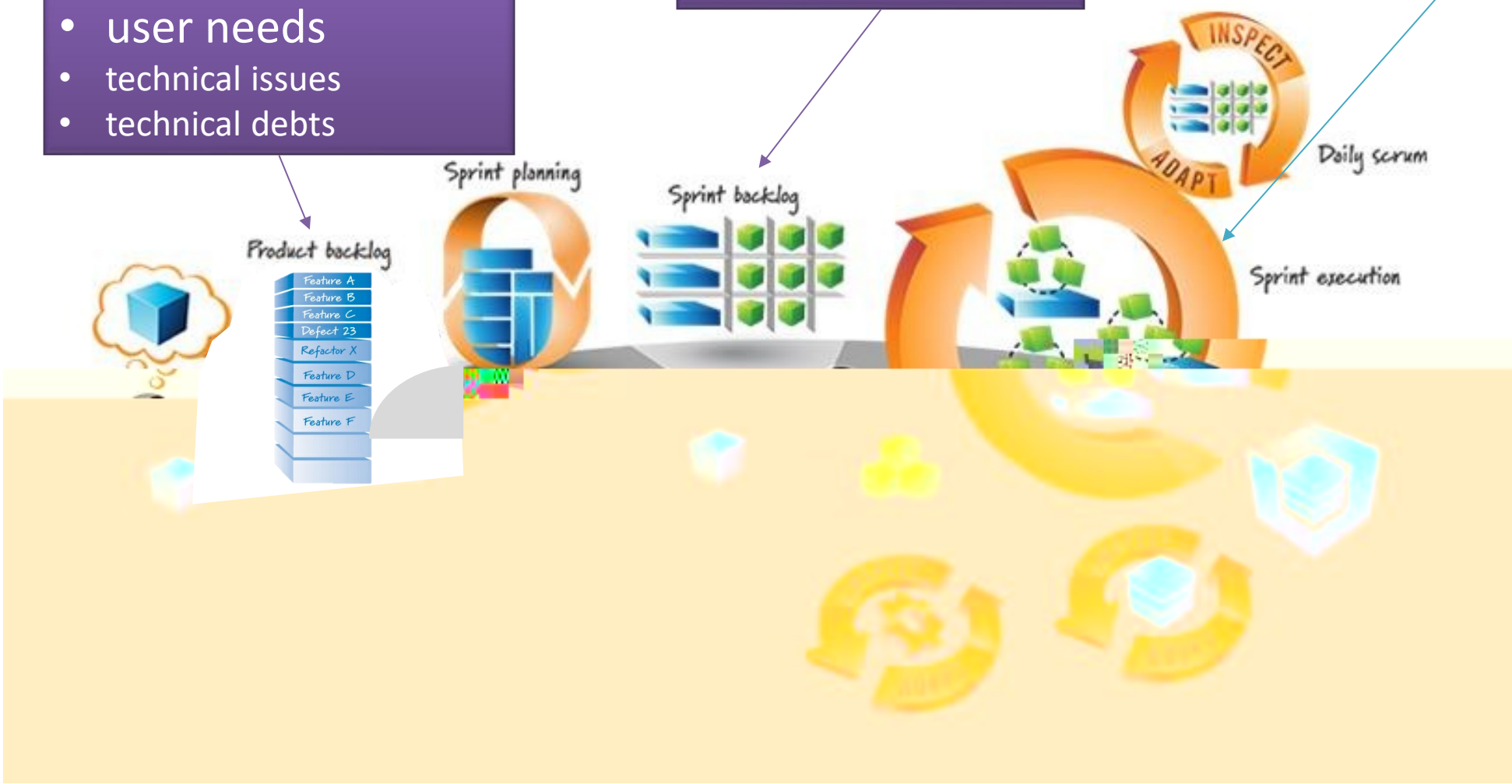
Looping on product backlog starting from top items till sprint capacity is reached

Overall Scrum Process

Artifact 1: Product backlog
A **totally ordered** list of
- user needs
- technical issues
- technical debts

Artifact 2: Sprint backlog
A **collection** of tasks

Good old style SW development
- no **prescribed** practices,
- we need some



Daily Scrum Meetings

One Meeting a Day Keeps the Other Meetings Away

- Why
  - to share the current status with the rest of the team
    - what did you do yesterday?
    - is anything in your way?
  - to jointly plan next day's work
    - what will you do today?
- When
  - every working day
    - it is the border between yesterday and today at whatever hour it takes place
  - no longer than 15 minutes
- Who
  - all team members, product owner, scrum master
  - any other stakeholder as silent spectator

Managing the sprint backlog

- Any team member can add, delete or change the sprint backlog
  - Nobody but team members can!
  - Work for the sprint emerges
- Individuals sign up for work of their own choosing
  - Work is never assigned

- Estimated work remaining is updated daily
  - If work is unclear, define a sprint backlog item with a larger amount of time and break it down later
  - Update work remaining as more becomes known

Tasks	Mon	Tues	Wed	Thurs	Fri
Code the user interface	8	4	8		
Code the middle tier	16	12	10	4	
Test the middle tier	8	12	16	11	8
Write online help	12				
Write the foo class	8	8	8	8	8
Add error logging			8	4	
Effort needed to complete	**52**	**40**	**50**	**27**	**16**

I will work with you

but we will need fresh data-sets

Yesterday I worked with Ben on task T666 but we did not finish it

we need help to understand the current architecture

Yesterday I replaced Ben on task T1024 and completed it with Anne. Then I started task T42 with Frank

Thanks, then in the morning we can help somebody on something else

no problems at all

Then let me start on T55, who wants to join me?

Today I'll work on T3 with Anna

Task T55 presents high business risks I would like to see it implemented soon



I will get them for you by 10 am

OK, then if it's ok for Anna, we will work on T11





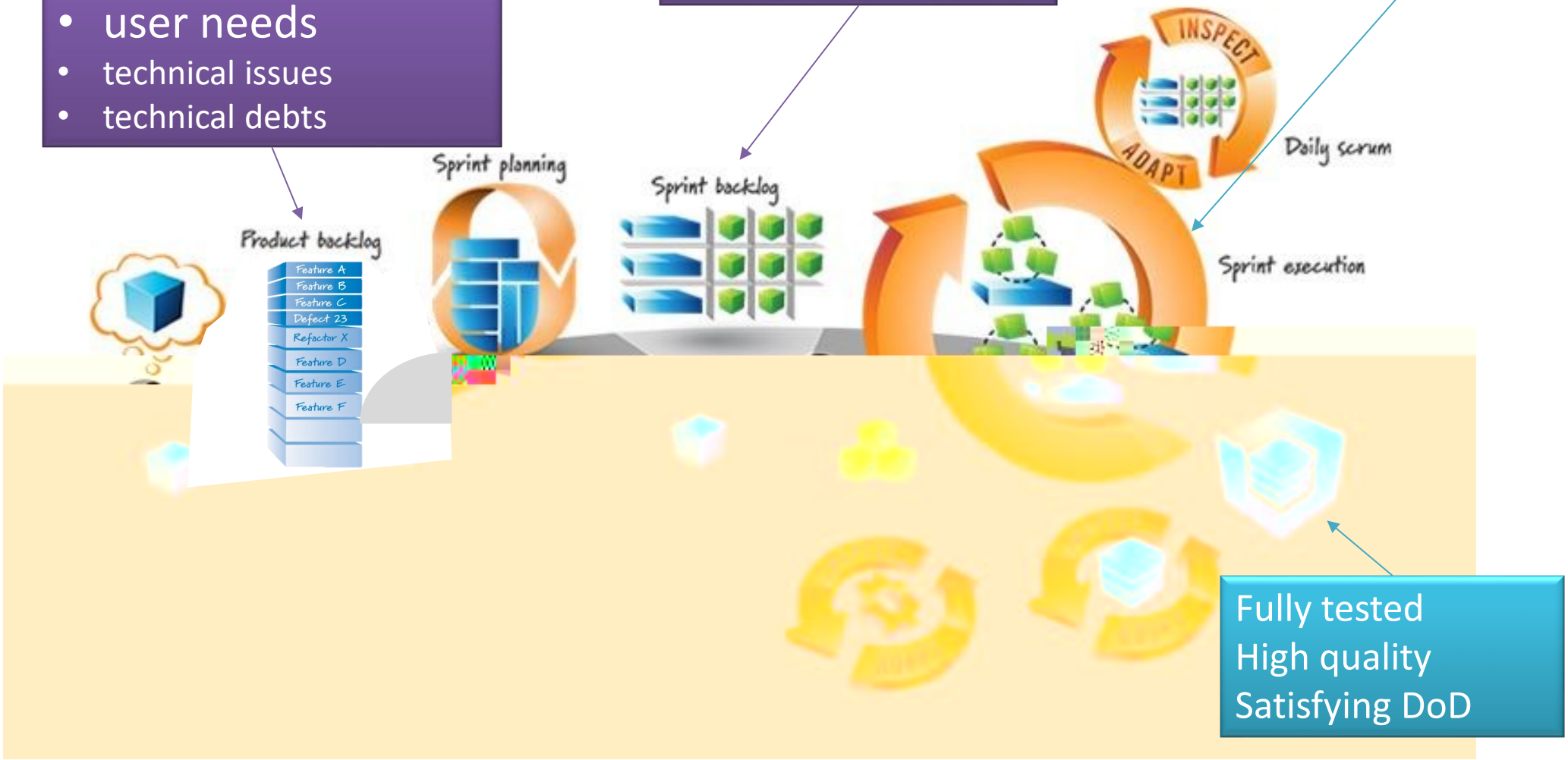
Dibris

Overall Scrum Process

Artifact 1: Product backlog
A **totally ordered** list of
- user needs
- technical issues
- technical debts

Artifact 2: Sprint backlog
A **collection** of tasks

Good old style SW development
- no **prescribed** practices,
- we need some



Fully tested
High quality
Satisfying DoD

The sprint review

- Inspection of the sprint product increment
  - demo of new features or underlying architecture
  - informal
    - 2-hour prep time rule
    - No slides
- Whole team *participates*
- Outcomes
  - Product owner approves/rejects
  - First feedback from stakeholders

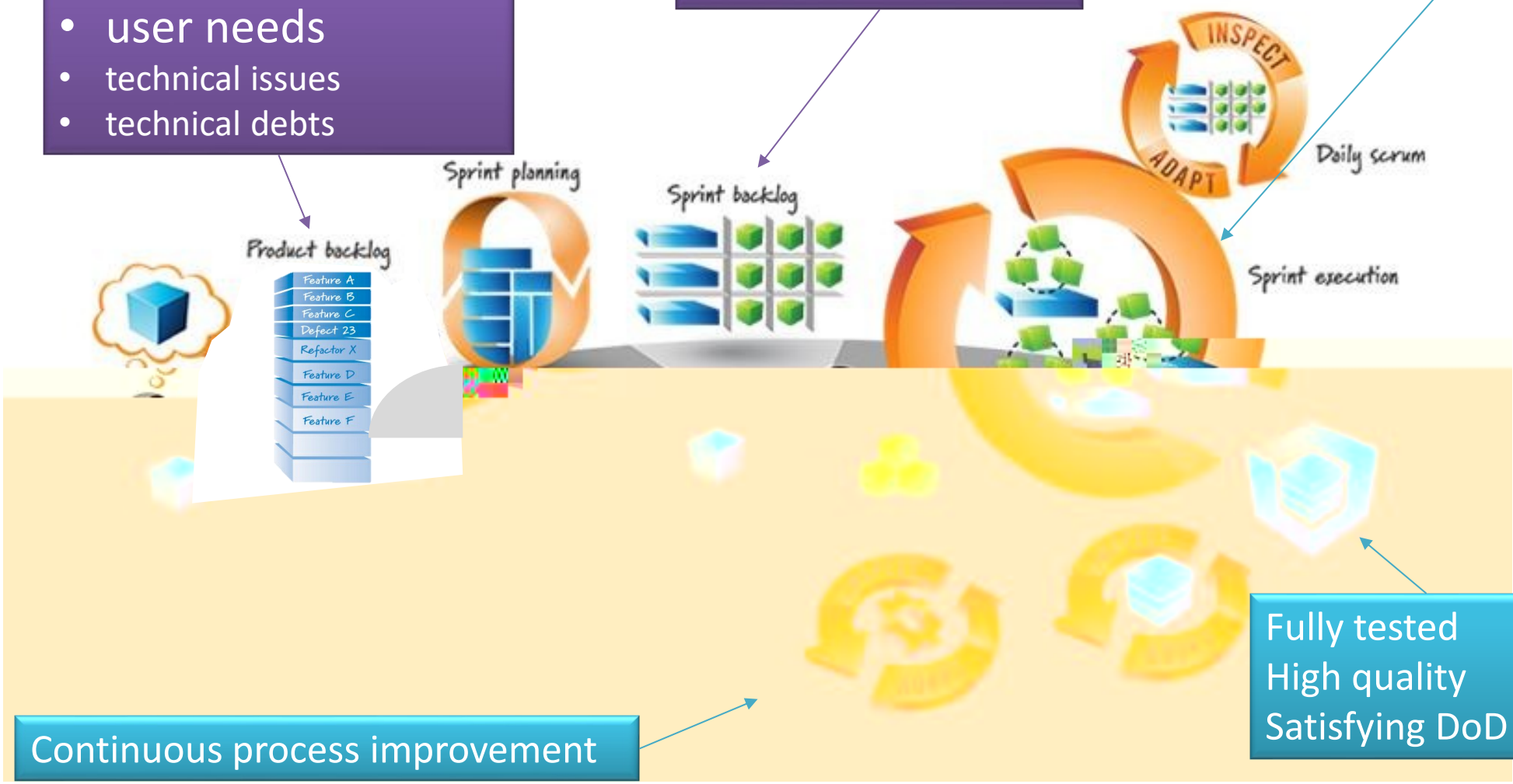


Overall Scrum Process

Artifact 1: Product backlog
A **totally ordered** list of
- user needs
- technical issues
- technical debts

Artifact 2: Sprint backlog
A **collection** of tasks

Good old style SW development
- no **prescribed** practices,
- we need some



Continuous process improvement

Fully tested
High quality
Satisfying DoD

Sprint retrospective

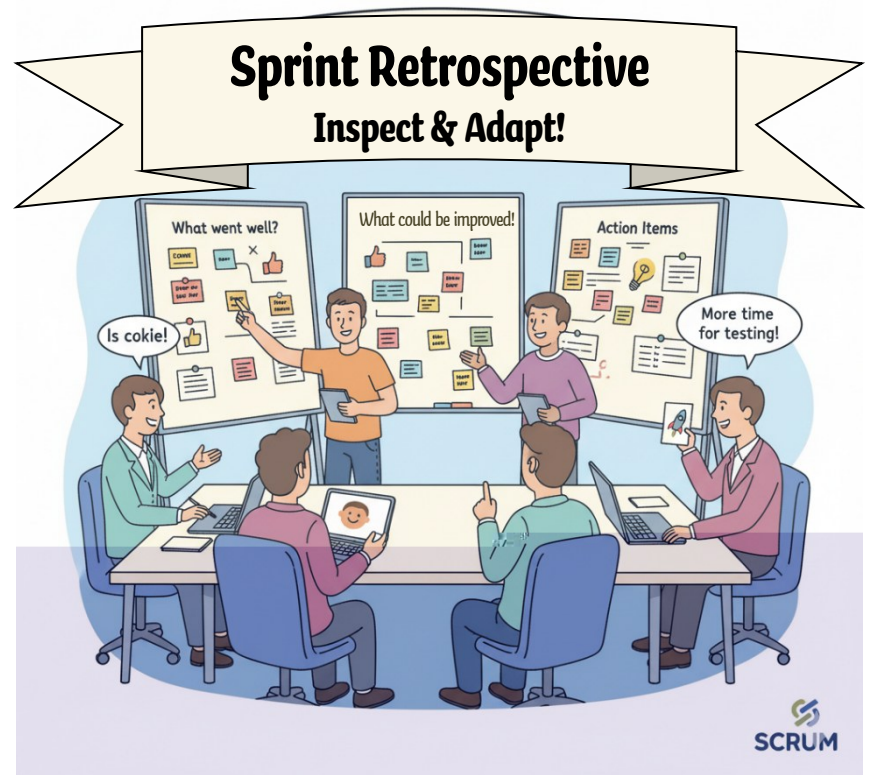
- Critique of development process
  - What worked
  - What did not work
  - New practices that could help
- Typically, 15–30 minutes
- Key roles *participate*
  - ScrumMaster
  - Product owner
  - Team
- Possibly others
  - Customer/Stakeholders
  - [Process] Consultants
  - “Apprentices”

Keep it!

Stop it!

Replacement

Give it a try!

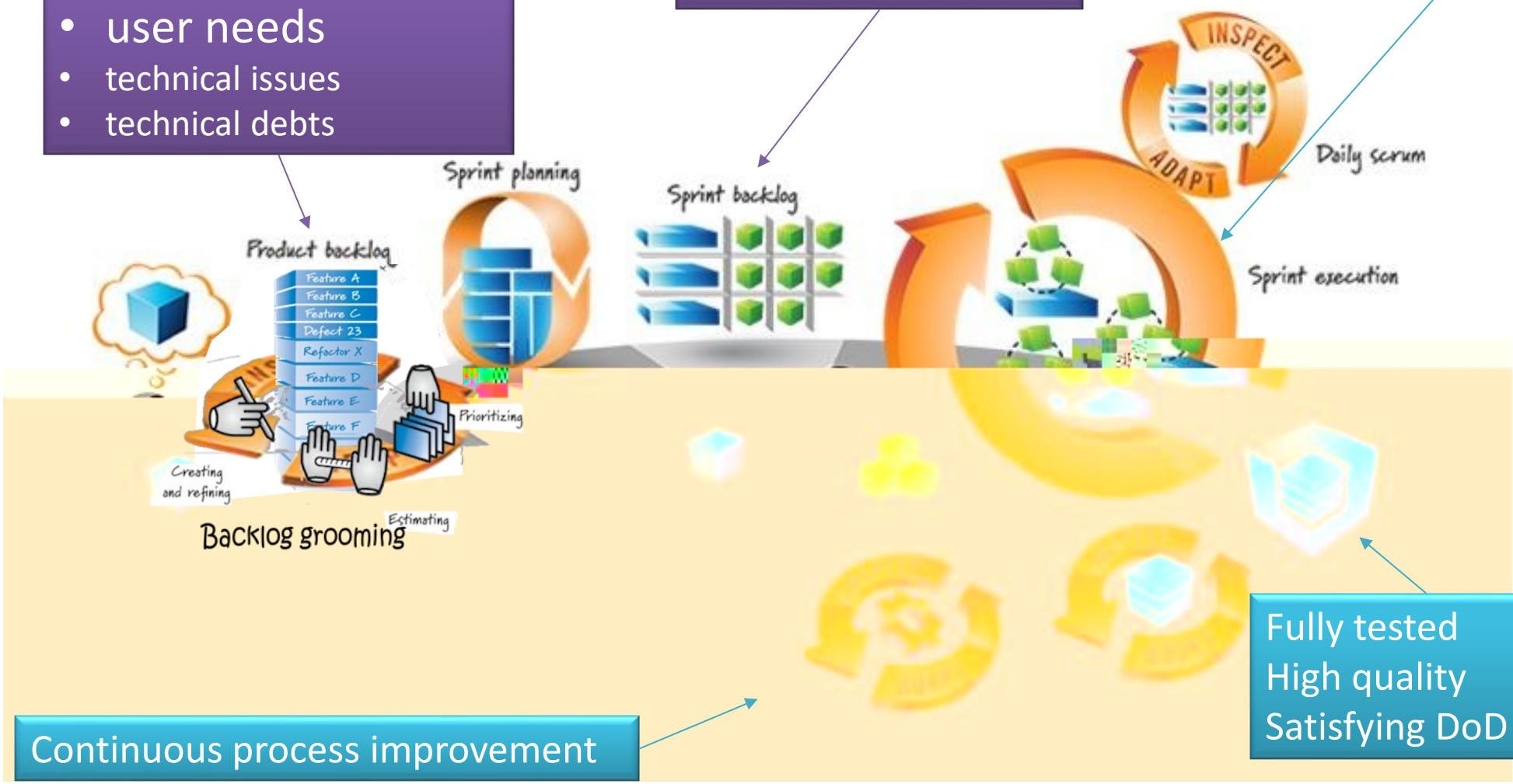


Overall Scrum Process

Artifact 1: Product backlog
A **totally ordered** list of
- user needs
- technical issues
- technical debts

Artifact 2: Sprint backlog
A **collection** of tasks

Good old style SW development
- no **prescribed** practices,
- we need some

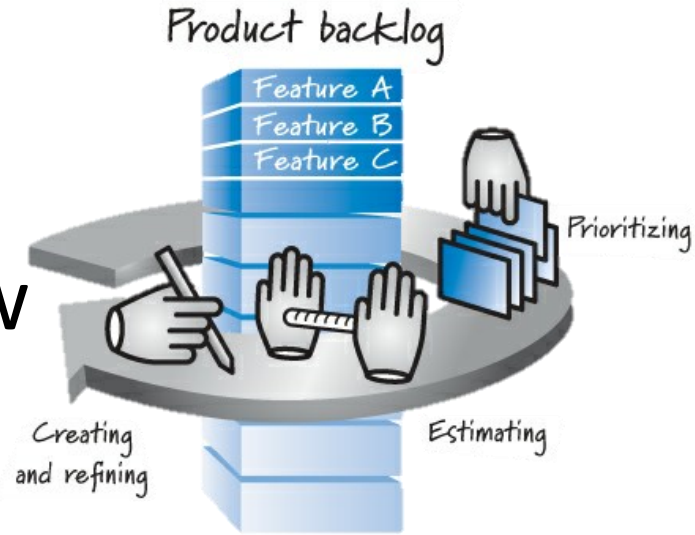


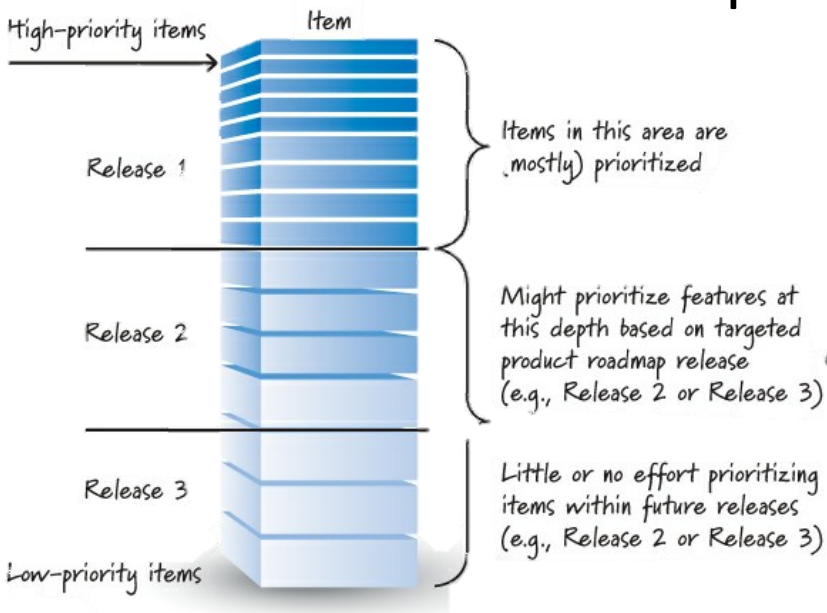
Continuous process improvement

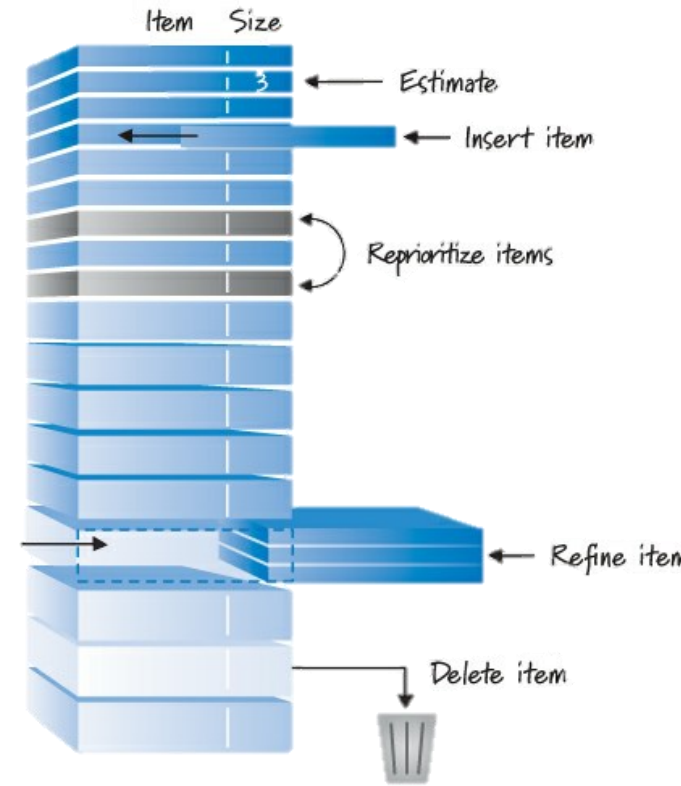
Fully tested
High quality
Satisfying DoD

Backlog grooming/refinement

- Last *ceremony* added
- Not clear position in the work flow
  - within sprint planning
  - within sprint review
  - standalone meeting
    - somewhere toward end of sprint







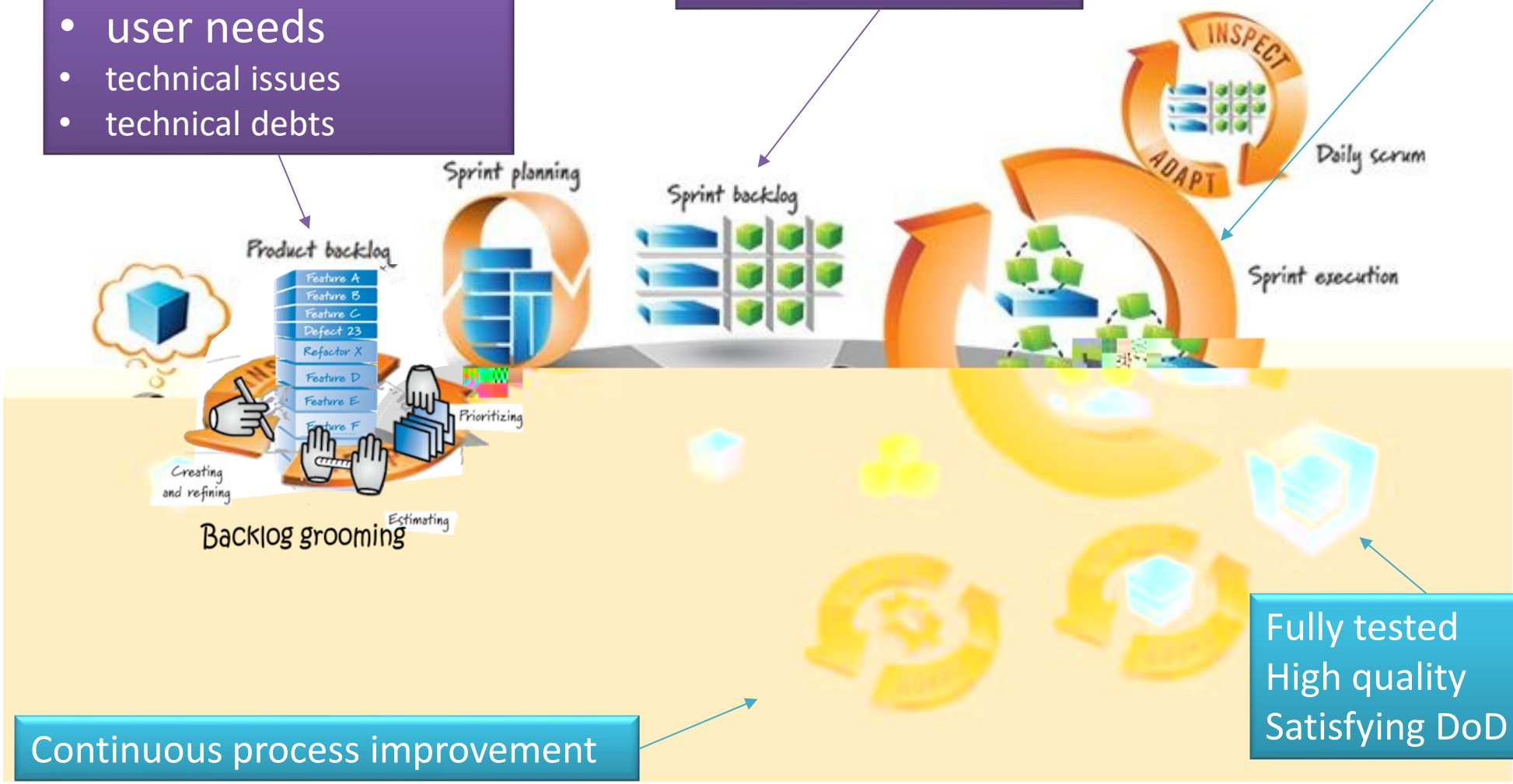
Dibris

Overall Scrum Process

Artifact 1: Product backlog
A **totally ordered** list of
- user needs
- technical issues
- technical debts

Artifact 2: Sprint backlog
A **collection** of tasks

Good old style SW development
- no **prescribed** practices,
- we need some



Continuous process improvement

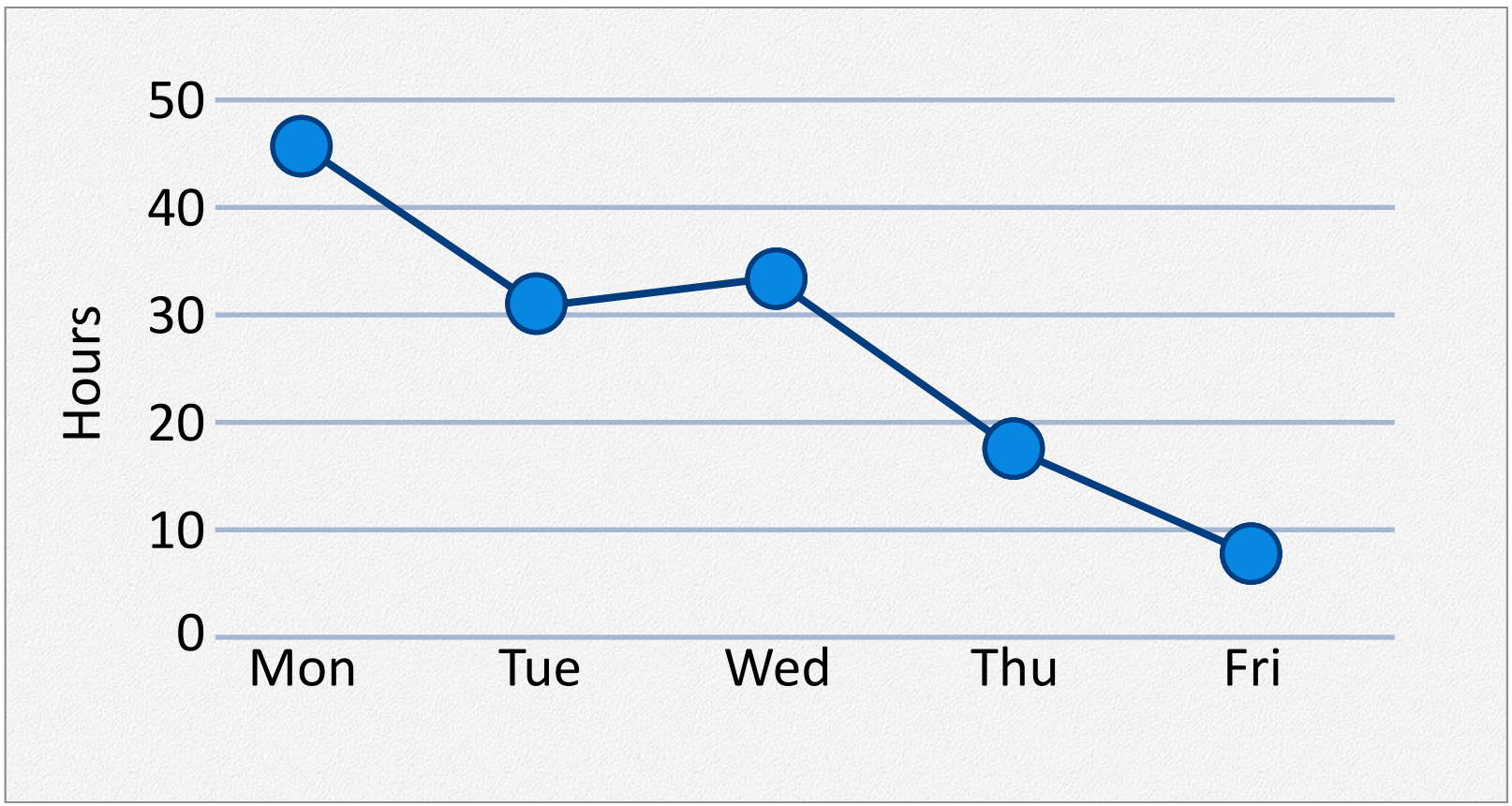
Fully tested
High quality
Satisfying DoD

All activities are time boxed

Dibris

A Sprint Burn Down Chart

Tasks	Mon	Tues	Wed	Thur	Fri
Code the user interface	8	4	8		
Code the middle tier	16	12	10	7	
Test the middle tier	8	16	16	11	8
Write online help	12				
	44	32	34	18	8



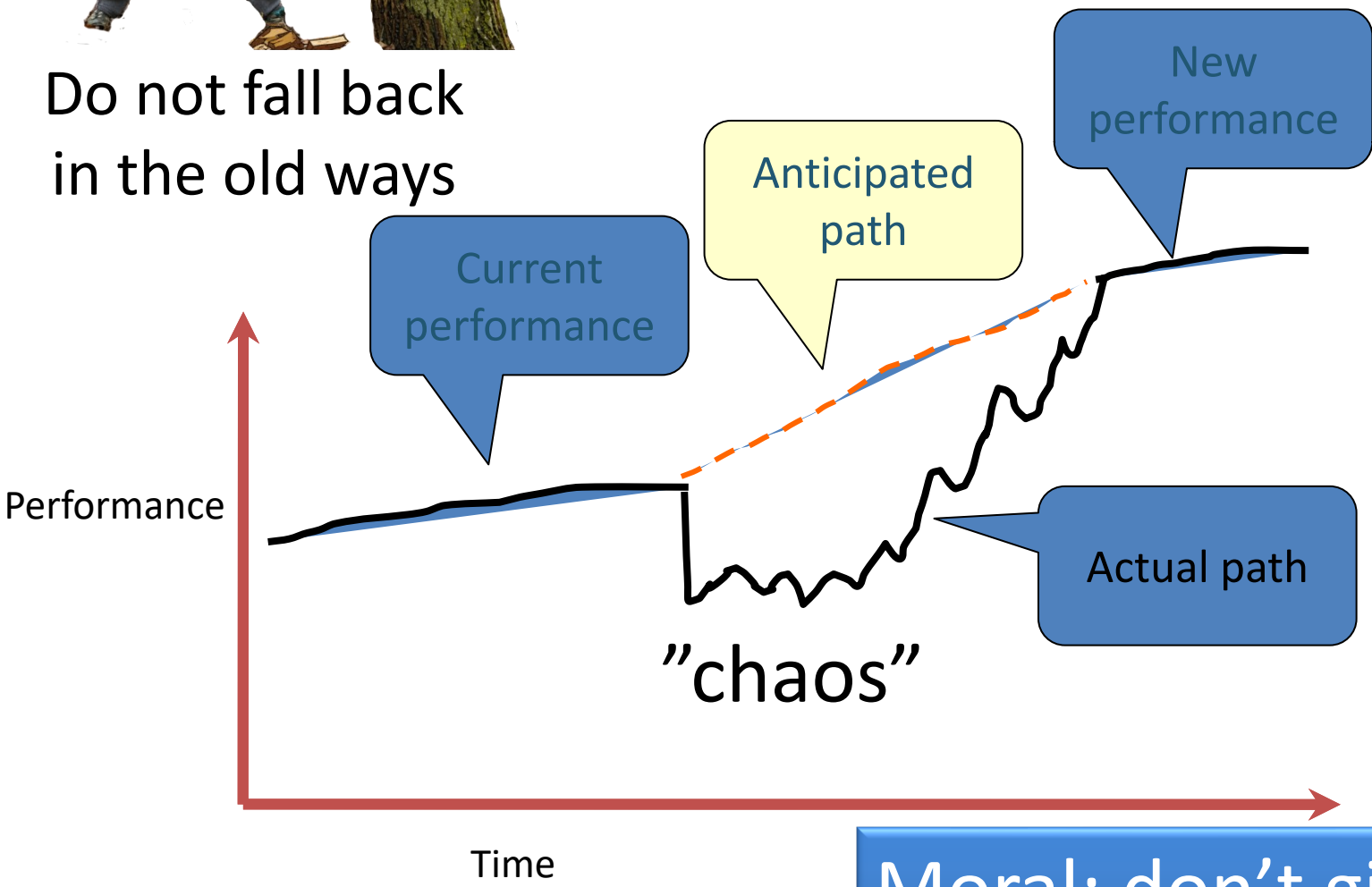
Dibris



Satir model

”the middle always looks like failure”

Do not fall back
in the old ways



Moral: don’t give up to the first difficulties

Dibris

OK, Let's do it!

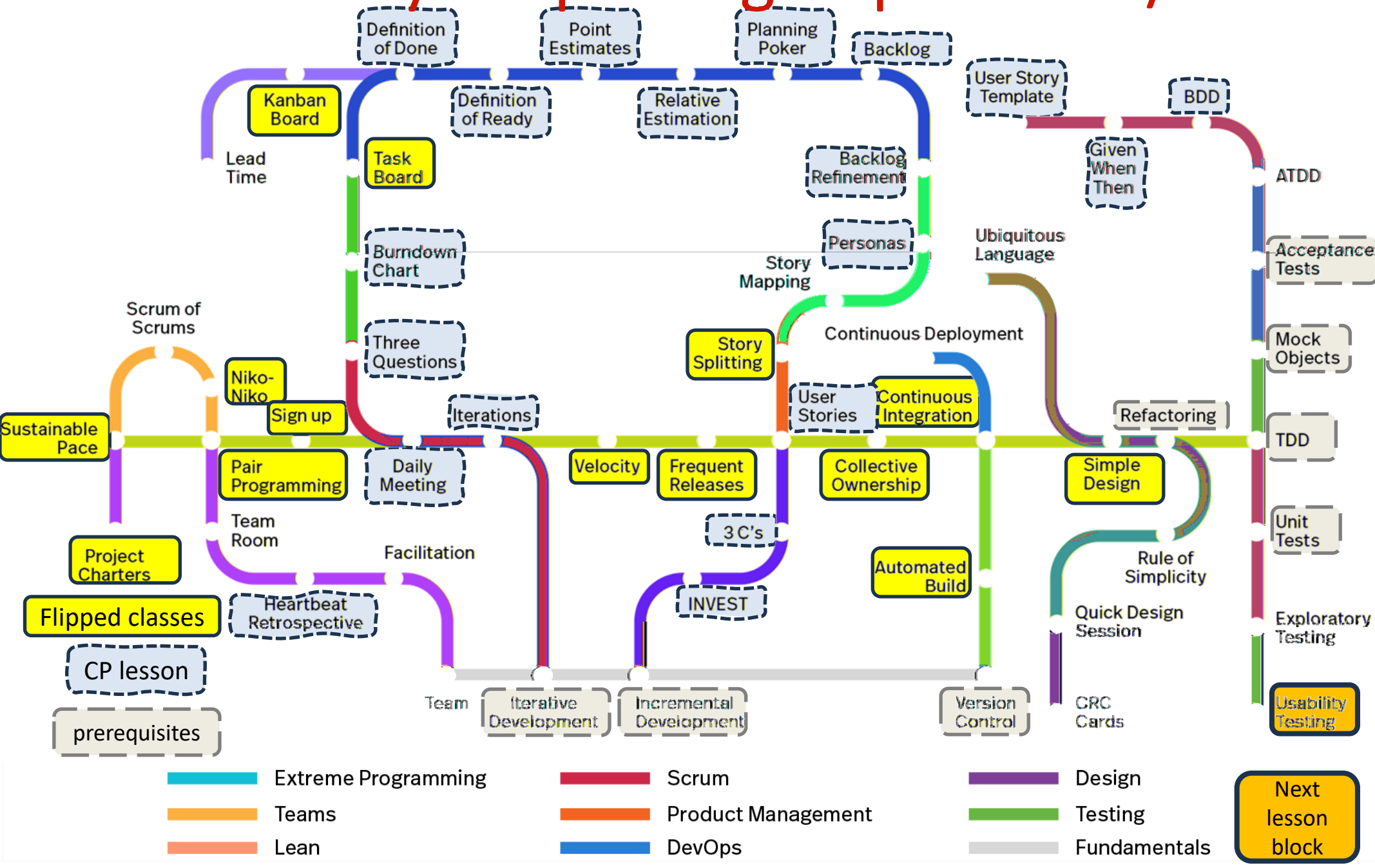
- First step to apply Scrum is….



- Problem
  Scrum is a high-level method
  - principles
  - overall process and milestones/ceremonies
- Missing low-level details
  - day-to-day work practices
    - just a bit here and there…daily meeting's 3 questions
  - choices of techniques and tools
    - not only for coding
    - just a bit here and there…
      burn down charts

Flipped lessons
Practices and techniques/tools
which we *could* adopt

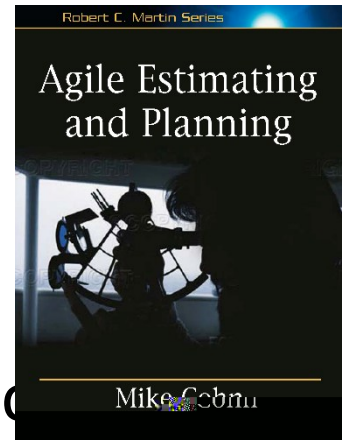
https://agilealliance.org/agile101/subway-map-to-agile-practices/



Acknowledgements

Slides based on

- An Introduction to Scrum
  by Mike Cohn
- Essential Scrum:
  A Practical Guide to the Most Po
  Agile Process
  by Kenneth S. Rubin
- Agile Testing lectures for Software QA Testing students
  Iana Mourza at Portnov Computer School
  - http://www.youtube.com/watch?v=AszbYDiygFo
  - http://www.youtube.com/watch?v=5E4O4NQCb68





Mountain Goat Software, LLC

